

Daylight Harvesting With Automatic Lighting

Imagine the availability of 60,000 units of light energy through an automatic system that monitors energy consumption to maximise efficiency. At the age of 48, Sekhar Nori, founder & CEO of Skyshade Daylights Pvt Ltd, teamed up with his colleagues to create an automated and integrated daylight harvesting system



SNEHA AMBASTHA

Sekhar Nori has come up with a product he calls the Day360 Integrated Daylight System (IDS). This is a lighting system that harvests day light to provide free lighting during the day and electrical lighting at night. It is controlled automatically by the system but has manual override option for those special occasions.

Sekhar Nori says, "During my former experience with solar energy field, where I worked with solar conversion, I found that the conversion ratio of solar energy is comparatively depressed. This caused the system costs to be relatively high. This is when we recognised that it is time to convey the visual solar energy as light itself."

In this system you do not convert day light into any other form of energy like heat or electricity. This helps increase the product's life, the product becomes more affordable which, in turn,

increases the return on investment.

What it is

It is a concept where both natural day light and electrical lighting are integrated into a single and simple platform. Depending on the availability of day light, the electrical lighting system switches on or off to complement light in the area.

Although the Day360 IDS was deduced from the idea of Lightpipe (a solar daylight system), it is a completely new solution with daylight system being just a part of it. The IDS makes use of electrical lighting, day lighting, light sensors and a lighting controller for an all-in-one system that even has smart meter functionality.

Sekhar says, "We did a lot of R&D for IDS for about one-and-a-half year. This year we finally finished its design and fabrication. IDS is a completely new way of providing an automatic lighting solution for commercial buildings."

How it works

The daylight system part of IDS has a light collector, a reflective system and light diffusers. The light collector collects sun light and the reflective system transports this light just like an optical fibre. The light diffusers take the light out of the tube and diffuse it into the building.

Lighting controllers. The lighting controllers take care of the entire working of the IDS and also help in switching between day light and electrical lighting. The lighting controller has three parts – daylight sensing, control system and a central control panel.

Sekhar says, "The light sensing element used by IDS is a special light sensor built by us. This is an open loop light sensor. Normally the light sensor used in building works in a closed loop. The light sensor that we build gives a continuous feedback to the lighting controller. In an open loop sensor the light is measured from the ambient sky and never measured in the task plane. Whereas in closed loop sensor light is continuously measured at the task plane (a task plane could be either floor level or any other plane) and is given as a feedback to the controller."

The light sensor continuously measures the ambient sun light. If the ambient sun light is less than the desired light intensity required in the room, the lighting controller turns on the electrical lighting. The electrical lighting is divided into three groups, each representing a step for switching on or off. Depending upon the light intensity, the controller turns on required number of steps in electric lighting. Nori adds, "The lighting controls and the central control panels are also built in-house. All lighting systems can be con-